for our community to come forward and adopt TB patients to help them get treatment for free. This is a one of its kind initiative in the world, which has grown into a people's movement. People from all walks of life are coming forward to help us end TB". The government is also enlisting TB Vijetas—people who have conquered the disease—for communication and community support strategies, which involve educating people, erasing stigma and extending emotional support to patients as well as their caregivers.

Diagnostics is a crucial frontier that the government is concentrating on. Unlike regular TB, it can be challenging to map the full extent of DR-TB because it requires specialised tests to detect drug resistance. These tests are often expensive and not widely available, particularly in low-resource settings. "The gold standard for identifying drug-resistant TB remains the conventional culture-based test," says Dr Bornali Dutta, director of respiratory medicine at Medanta – The Medicity, Gurugram. "But since the TB bacteria grows slowly, it can take two weeks before results are known." And while new testing technology, such as TrueNAT/ CBNAAT, are game-changers in the rapid and accurate diagnosis of TB and in detecting rifampicin resistance, they are difficult and slow to develop.

"The ideal diagnostic tool for detection of DR-TB should be rapid, accurate, cost-effective, have a long shelf life and provide comprehensive coverage of all ATT drugs [antitubercular agents are a group of drugs used to treat TB]. The challenge is to amalgamate all the above in a single test," says Dr Gunisha Pasricha, principal scientist and infectious diseases expert at MedGenome Labs in Delhi. She and her team have come out with a test called SPIT SEQ, which can not only detect DR-TB in 14 days



"THE IDEAL DIAGNOSTIC TOOL FOR DR-TB DETECTION SHOULD BE **RAPID, ACCURATE, COST-EFFECTIVE, HAVE A LONG SHELF LIFE AND PROVIDE COMPREHENSIVE COVERAGE** OF ALL ANTITUBERCULAR DRUGS. THE CHALLENGE IS TO AMALGAMATE ALL THE ABOVE IN A SINGLE TEST"

–DR GUNISHA PASRICHA,
Principal scientist, MedGenome Labs



"INDIA IS INSPIRING THE WORLD IN ELIMINATING TB. **EVERY COUNTRY SHOULD HAVE A NI-KSHAY MITRA-LIKE** INITIATIVE.... IT'S HAPPENING NOT IN PILOT MODE BUT AT SCALE"

–LUCICA DITIU, *Executive Director, Stop TB Partnership*

but also describe the mutation and associated resistance for 18 drugs, including first line, second line and new drugs. The test has been found to have 100 per cent sensitivity and 98.04 per cent specificity. The advent of newer technology means a quick molecular test can find out whether DR-TB is present and which drugs it is resistant to, thus helping start treatment quickly. The new tests also help determine the two things essential for TB treatment—the required duration and dose of the medicines.

SPEEDING UP DIAGNOSIS

To widen the reach of these technologies, the government has been ploughing money into human resources, infrastructure and policy. The number of TrueNAT/ CBNAAT facilities in the country has gone up to 5,090 from a mere 40 in 2014. As of now, there are 80 labs in the country that support the liquid culture system for DR-TB, which has now been expanded to include resistance to more drugs such as linezolid and pyrazinamide.

Significant ground deployment has helped these efforts. In 2021, all of 34 states/UTs conducted door-to-door campaigns, under which over 223 million people were screened, 1,752,903 samples were tested, and 73,772 additional TB patients were identified. Eighty-one mobile TB diagnostic vans have been made available to states to conduct diagnoses in remote areas. In a hugely positive move, the government has also begun treating close household contacts of TB patients and getting them tested. Even if they have latent TB (LTI), they can be treated to ensure that it doesn't develop into active disease. The Mantoux test, which involves injecting a substance called PPD tuberculin into the skin, can help determine LTI in the body. It's available at most private hospitals now for an average cost of Rs 100.

The results have begun to show. In 2021, there was a 19 per cent increase in the number of overall TB cases diag-